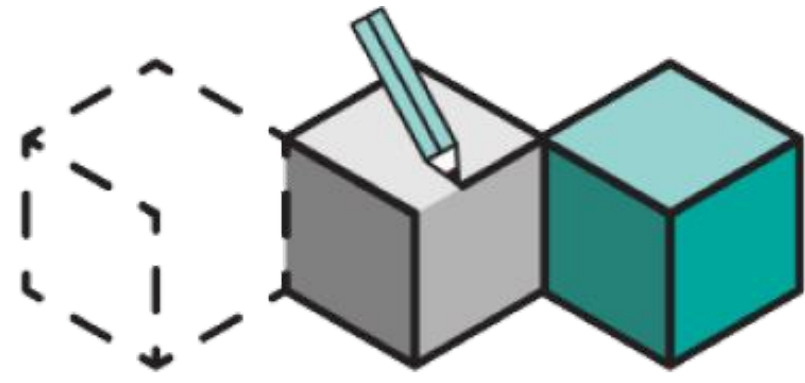




NYU Prototyping Fund
designlab @ MakerSpace

Spring 2026

KendoSense

Wearable Motion Analytics for Kendo Training

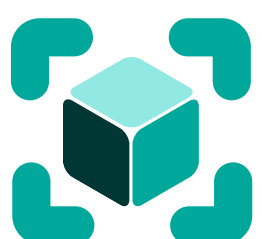
Project Description:

Kendo Practitioners rely heavily on subjective feedback and manual video review to evaluate technique. However, key elements such as strike timing, grip mechanics, and reaction speed occur within milliseconds and are difficult to quantify. KendoSense is a wearable motion sensing system designed to provide objective feedback during training. The system uses an onboard inertial measurement unit (IMU) embedded within the tsuba (hand guard) of a shinai (bamboo sword) to capture acceleration and rotational data during strikes. Sensor data is recorded and analyzed to identify motion patterns associated with strike execution. This project explores how embedded sensing and data logging can be applied to high-speed, real-world athletic environments where traditional tracking methods are limited. A threshold-based method was implemented to detect strike events from angular velocity data.

Top Lessons:

This project highlighted that hardware integration is often more challenging than software development, as mechanical fit, wiring stability, and power management introduce significant complexity. It also emphasized the importance of understanding raw sensor data before developing algorithms, as observing IMU signals directly is critical for identifying meaningful motion patterns. Additionally, simplifying the system design proved to be highly effective; reducing unnecessary components improved both reliability and development speed. Finally, working within real-world constraints, such as high-speed motion, physical impact, and wearable limitations, demonstrated that practical consideration plays a major role in engineering design decisions.

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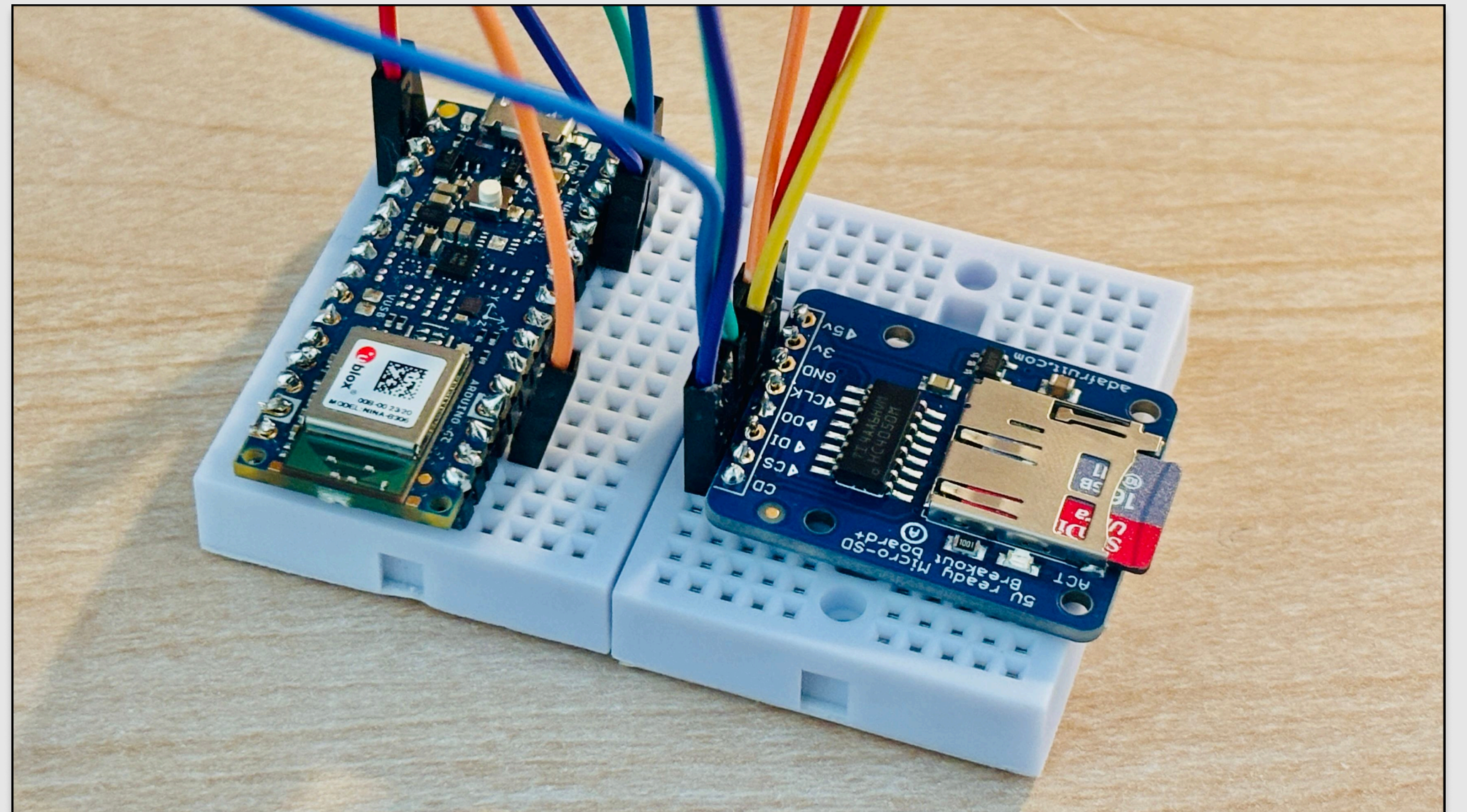


Figure 1. Prototype of the KendoSense system, consisting of an Arduino Nano 33 BLE Sense and microSD module integrated into a custom tsuba enclosure for real-time sensing and data logging.

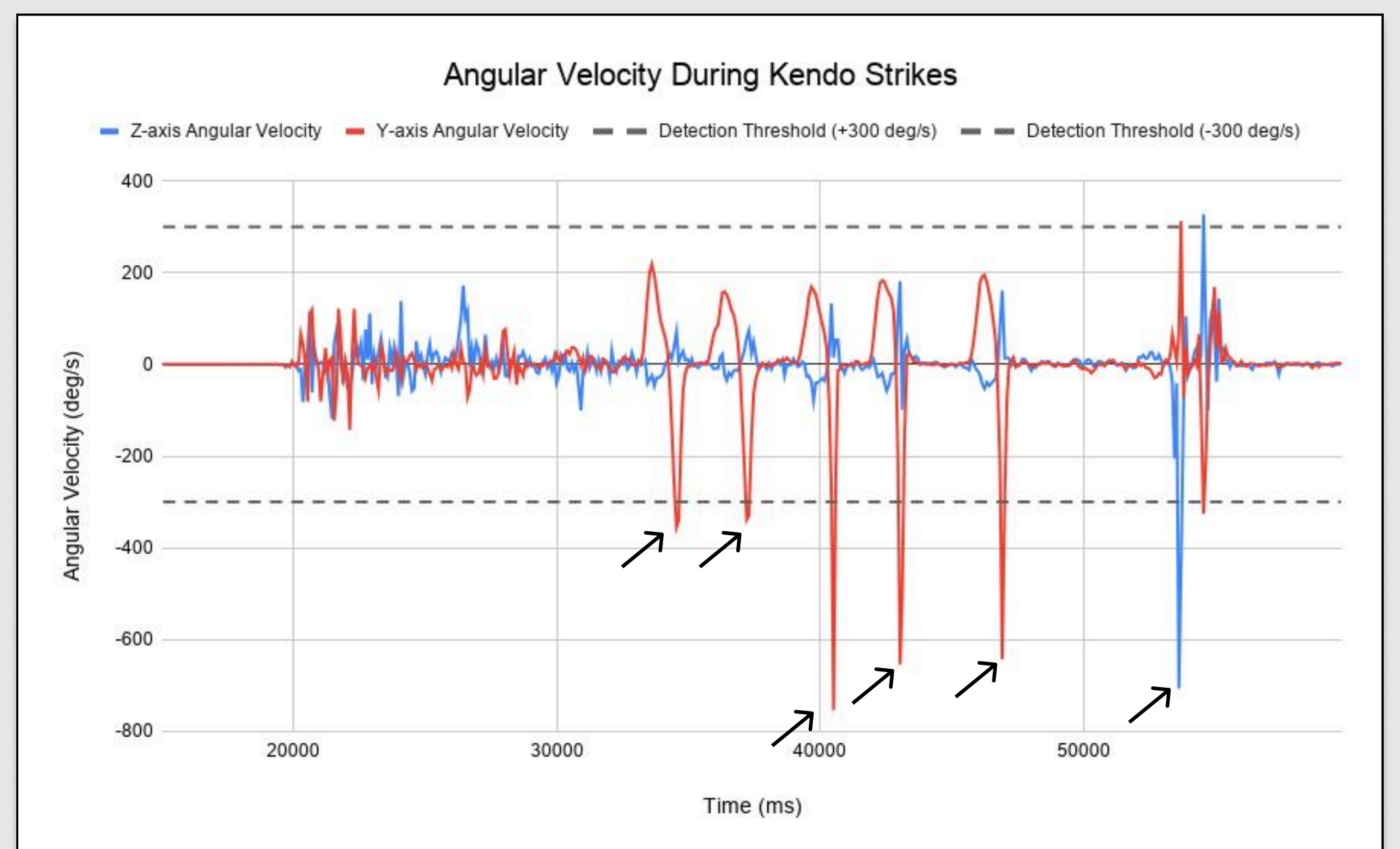


Figure 2. Angular velocity measured by the onboard IMU during kendo strikes. Clear spikes exceeding ± 300 deg/s correspond to intentional strike motions, enabling threshold-based detection.

How we built it:

The system was built using an Arduino Nano 33 BLE Sense, which includes an onboard inertial measurement unit (IMU) for capturing motion data. A micro-SD module was integrated to enable onboard data logging, allowing sensor readings to be stored in CSV format for later analysis. To support wearable deployment, a custom 3D-modeled tsuba enclosure was designed to house the electronics in a compact and protective form factor. The data collection pipeline was implemented such that IMU data is captured by the Arduino, timestamped, and written directly to the SD card. Initial firmware was developed to record acceleration and gyroscope values in real time, enabling analysis of motion patterns during simulated strike movements. The system was iteratively tested and refined through repeated trials to ensure stable data collection. In addition, a portable power system using a LiPo Battery and a 3.3V voltage regulator was planned to support untethered operation in real training environments.